

# Clinical efficacy and safety of drug interventions for primary and secondary prevention of osteoporotic fractures in postmenopausal women: a network meta-analysis followed by a factor and cluster analysis

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## Citation

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## Review question

To evaluate the comparative efficacy and safety of fifteen anti-osteoporotic drug interventions for postmenopausal osteoporosis respectively in the primary prevention group and in the secondary prevention group, and if possible to identify the optimal treatment(s) when efficacy and safety both considered respectively for primary prevention and for secondary prevention of osteoporotic fractures in postmenopausal women.

## Searches

We will systematically search PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) for English-language articles without sample size restrictions.

Search strategies will be developed using text words (e.g., "osteoporosis", "postmenopausal women", "postmenopausal osteoporosis", "Romosozumab", "Abaloparatide", "Parathyroid Hormone (1-84)", "PTH (1-84)", "PTH", "fracture\*", and "randomized controlled trial"), and medical subject headings (MeSH; e.g., "Osteoporosis, Postmenopausal", "Denosumab", and "Parathyroid Hormone") or medical supplementary concepts (e.g., "Abaloparatide", and "AMG 785").

In addition, Google Scholar and the references in systematic reviews previously published will be also used to search for relevant primary studies.

The searches will be re-run to search for more studies possibly eligible for inclusion just before the final analyses.

## Types of study to be included

Randomized controlled trials (RCTs) lasting for at least one year and comparing an anti-osteoporosis agent with one or more other anti-osteoporosis agents, or with placebo. Calcium and/or vitamin D given to participants in both the intervention and comparator arms will be allowed.

## Condition or domain being studied

Osteoporosis, with the overwhelming majority occurring in postmenopausal women (PMW), leads to 1.5 million osteoporotic fractures and a total annual cost of \$20 billion per year in the USA. Meanwhile, osteoporotic fractures are associated with an increased risk of death, a loss of self-esteem, and other adverse outcomes. Moreover, the number of women at risk for osteoporotic fractures and the relevant medical costs in the next decades are expected to continue to rise. The vast majority of fragility fractures occur in osteopenic women (i.e., those with a BMD T-score between -1 and -2.5), which reveals the importance of primary prevention of osteoporotic fractures in PMW. Meanwhile, women who have developed a fragility fracture are at higher risk for secondary fractures, which reveals the importance of secondary prevention. However, the comparative efficacy and safety of different osteoporosis drugs respectively used for primary prevention and secondary prevention is unknown. To make informed decisions on the selection of anti-osteoporotic agents in clinical reality, the clinicians and susceptible individuals need to know the comparative efficacy and safety of various drug interventions and the best intervention(s) with efficacy and

safety both considered, respectively for primary and secondary prevention.

### Participants/population

Postmenopausal women with primary osteoporosis or osteopenia.

### Intervention(s), exposure(s)

Fifteen active drug interventions of interest: abaloparatide (subcutaneous, 80 µg/day), teriparatide (subcutaneous, 20 µg/day), romosozumab (subcutaneous, 210 mg/month), denosumab (subcutaneous, 60 mg/6 months), parathyroid hormone (1-84) (subcutaneous, 100 µg/day), zoledronate (intravenous, 5 mg, once per 24 months), zoledronate (intravenous, 5 mg, once per 18 months), zoledronate (intravenous, 5 mg, once per 12 months), strontium ranelate (oral, 2 g/day), lasofoxifene (oral, 0.5 mg/day), bazedoxifene (oral, 20 mg/day), raloxifene (oral, 60 mg/day), risedronate (oral, 35 mg/week or 5 mg/day), alendronate (oral, 35 or 70 mg/week, or 5 or 10 mg/day), and ibandronate (oral, 150 mg/month).

### Comparator(s)/control

One of the 15 active drug interventions of interest or placebo.

### Main outcome(s)

Efficacy outcomes:

Vertebral fractures;

Nonvertebral fractures.

Safety outcomes:

Tolerability, namely, withdrawals due to adverse events;

Acceptability, namely, withdrawals due to any cause.

#### \* Measures of effect

Not applicable.

### Additional outcome(s)

None.

#### \* Measures of effect

Not applicable.

### Data extraction (selection and coding)

From each eligible trial two authors will independently collect the following data of interest using a standard data extraction form created in Microsoft Excel 2013: (1) study characteristics (e.g., publication year, and quality score); (2) participant characteristics (e.g., mean age, prevalent fracture proportion, and bone mineral density T score at the femoral neck measured by DXA); (3) intervention characteristics (e.g., type, and duration); and (4) outcome data (study-level survival data, namely hazard ratio (HR) and its 95% confidence interval (CI), for nonvertebral fracture; and study-level dichotomous data, namely event number and total number, for the three other outcomes). Discussion between them or the involvement of a third author will address possible disagreements.

### Risk of bias (quality) assessment

Included RCTs will be assessed for quality by two independent authors according to the Jadad scale of which the final quality score is an integer in the range of 0 to 5. The Jadad total score is an additive combination of the score of randomization, the score of double blinding, and the score of withdrawals and

dropouts. In terms of randomization, a trial which is described as randomized and is with appropriate randomization method obtains a score of 2; a trial which is described as randomized but is not with detailed randomization method obtains a score of 1; and a trial with no mention of randomization obtains a score of 0. With the similar method, a trial in aspect of double blinding is graded. With respect to withdrawals and dropouts, if there is a detailed description about withdrawals and dropouts a trial will obtain a score of 1, otherwise the trial will obtain a score of 0.

### Strategy for data synthesis

Bayesian random-effects network meta-analyses will be conducted for two efficacy outcomes (vertebral fractures and nonvertebral fractures) and two safety outcomes (tolerability and acceptability) respectively in the primary prevention group and in the secondary prevention group. We will estimate hazard ratios (HRs) and 95% confidence intervals (CIs) for nonvertebral fractures, and estimate risk ratios (RRs) and 95% CIs for the three other outcomes. We will compute  $I^2$  statistic for heterogeneity assessment, and build node-splitting model for inconsistency assessment. When possible, factor and cluster analyses based on the surface under the cumulative ranking curve (SUCRA) values deriving from network meta-analyses will be conducted to identify the best drug intervention(s) with efficacy and safety both considered. Network meta-analyses will be done using JAGS (version 4.3.0) and R (version 3.6.0), factor analyses will be done using SPSS (version 24.0), and network plots and clustered ranking plots will be plotted using Stata (version 15.1).

### Analysis of subgroups or subsets

We will do subgroup network meta-analyses stratified by the primary prevention data and the secondary prevention data.

### Contact details for further information

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### Organisational affiliation of the review

Kunming Medical University

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### Type and method of review

Intervention, Meta-analysis, Network meta-analysis, Prevention, Systematic review

### Anticipated or actual start date

27 January 2019

### Anticipated completion date

20 October 2019

### Funding sources/sponsors

None

### Conflicts of interest

None known

### Language

English

### Country

China

## Stage of review

Review Ongoing

## Subject index terms status

Subject indexing assigned by CRD

## Subject index terms

Female; Fractures, Bone; Humans; Network Meta-Analysis; Osteoporosis; Postmenopause

## Date of registration in PROSPERO

22 August 2019

## Date of publication of this version

17 February 2020

## Stage of review at time of this submission

| Stage                                                           | Started | Completed |
|-----------------------------------------------------------------|---------|-----------|
| Preliminary searches                                            | Yes     | No        |
| Piloting of the study selection process                         | Yes     | No        |
| Formal screening of search results against eligibility criteria | No      | No        |
| Data extraction                                                 | No      | No        |
| Risk of bias (quality) assessment                               | No      | No        |
| Data analysis                                                   | No      | No        |

*The record owner confirms that the information they have supplied for this submission is accurate and complete and they understand that deliberate provision of inaccurate information or omission of data may be construed as scientific misconduct.*

*The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.*

## Versions

22 August 2019

17 February 2020

### PROSPERO

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